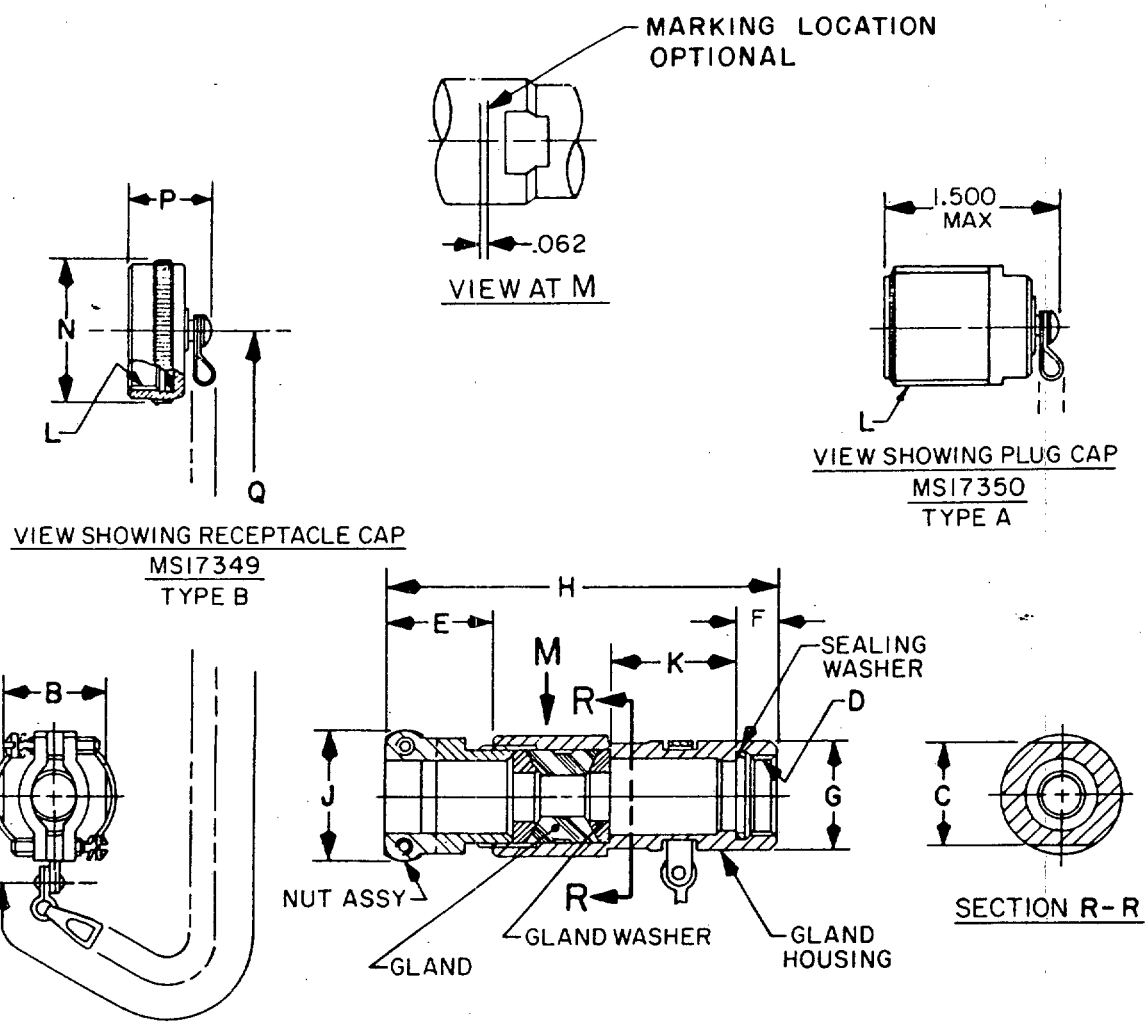
MIL-C-85049/4
16 January 1980
SUPERSEDING
MS17342B
1 June 1966, and
MS90569
25 May 1967

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, CABLE SEALING, STRAIGHT, STEP-UP, CATEGORY 1A (FOR MIL-C-22992 CONNECTORS, CLASSES C, J, AND R)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the accessories described herein shall consist of this document and the latest issue of Specification MIL-C-85049.



Configuration A. Style 1 cable sealing.

FIGURE 1. Configurations and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

SIZE NO.	SHELL SIZE	CABLE RANGE MAX DIA	CABLE RANGE MIN DIA	B +.000 -.010	C +.000 -.010	D THREAD CLASS 2B-LH	E FREE LG MAX	F ±.016	G DIA +.010 -.020	H MAX	J ±.031	K +.015 -.025	L THREAD (PLATED) CLASS 2 (A OR B)	N DIA MAX	P MAX	Q WITHIN ONE LINK
1	12	0.530	0.436	1.000	0.812	0.7500-20UNEF	1.062	.479	0.938	4.275	1.375	1.219	0.8750-0.1P-0.2L-DS	1.094	0.765	5.00
2	12	.500	.406	0.875	.938	.7500-20UNEF	0.969		.938	3.678	1.125	0.871	.8750-0.1P-0.2L-DS	1.094		
3	12	.405	.311	1.000	.812	.7500-20UNEF	1.062		.938	4.275	1.375	1.219	.8750-0.1P-0.2L-DS	1.094		
4	14	.605	.511	1.000	1.062	.8750-20UNEF			1.062	4.177	1.375	1.121	1.0000-0.1P-0.2L-DS	1.219		
5		.625	.531	1.062	1.125					4.489	1.562	1.371				
6		.530	.436	1.000	1.062					4.177	1.375	1.121				
7	14	.405	.311	1.000	1.062	.8750-20UNEF	1.062		1.062	4.177	1.375	1.121	1.0000-0.1P-0.2L-DS	1.219	.765	
8	18	.828	.715	1.188	1.250	1.1250-18UNEF	1.094		1.312	4.584	1.688	1.343	1.2500-0.1P-0.2L-DS	1.469	.980	
9	20	.750	.637	1.312	1.375	1.2500-18UNEF	1.094		1.438	4.647	1.812	1.371	1.3750-0.1P-0.2L-DS	1.562		
10		1.000	.875	1.546			1.281			4.896	2.125					
11		0.900	.787	1.312			1.094			4.647	1.812					
12	20	1.055	.930	1.546	1.375	1.2500-18NEF	1.281		1.438	4.896	2.125		1.3750-0.1P-0.2L-DS	1.562		5.00
13	22	1.000	.875	1.546	1.625	1.3750-18UNEF			1.562	4.896	2.125		1.5000-0.1P-0.2L-DS	1.688		6.00
14	22	1.109	.984	1.546	1.625	1.3750-18UNEF			1.562	4.896	2.125	1.371	1.5000-0.1P-0.2L-DS	1.688		
15	24	1.180	1.055	1.780	1.875	1.6250-18UNEF			1.812	4.959	2.469	1.372	1.7500-0.1P-0.2L-DS	1.938		
16		1.310	1.185	1.780							2.469	1.372				
17		1.230	1.105	1.780							2.469	1.372				
18	24	1.375	1.250	2.000	1.875	1.6250-18UNEF			1.812	4.959	2.625	1.309	1.7500-0.1P-0.2L-DS	1.938		
19	28	1.445	1.320		2.062	1.8750-16UN			2.062	5.021		1.371	2.0000-0.1P-0.2L-DS	2.219		
20	28	1.531	1.406		2.062	1.8750-16UN			2.062	5.021		1.371	2.0000-0.1P-0.2L-DS	2.219		
21	28	1.375	1.250	2.000	2.062	1.8750-16UN			2.062	5.021	2.625	1.371	2.0000-0.1P-0.2L-DS	2.219		
22	32	1.656	1.531	2.250	2.312	2.0625-16UNS	1.281		2.312	5.083	2.953	1.370	2.2500-0.1P-0.2L-DS	2.469		
23		1.828	1.700	2.438			1.391			5.385	3.171	1.375				
24		1.562	1.437	2.250			1.281			5.083	2.953	1.370				
25	32	1.730	1.605	2.438	2.312	2.0625-16UNS	1.391		2.312	5.385	3.171	1.375	2.2500-0.1P-0.2L-DS	2.469		
26	36	1.900	1.775	2.438	2.500	2.3125-16UNS	1.391		2.562	5.354	3.171	1.344	2.5000-0.1P-0.2L-DS	2.719		
27	36	1.730	1.605	2.438	2.500	2.3125-16UNS	1.391	.479	2.562	5.354	3.171	1.344	2.5000-0.1P-0.2L-DS	2.719	.980	6.00

Configuration A. Style 1 cable sealing - Continued.

FIGURE 1. Configurations and dimensions - Continued.

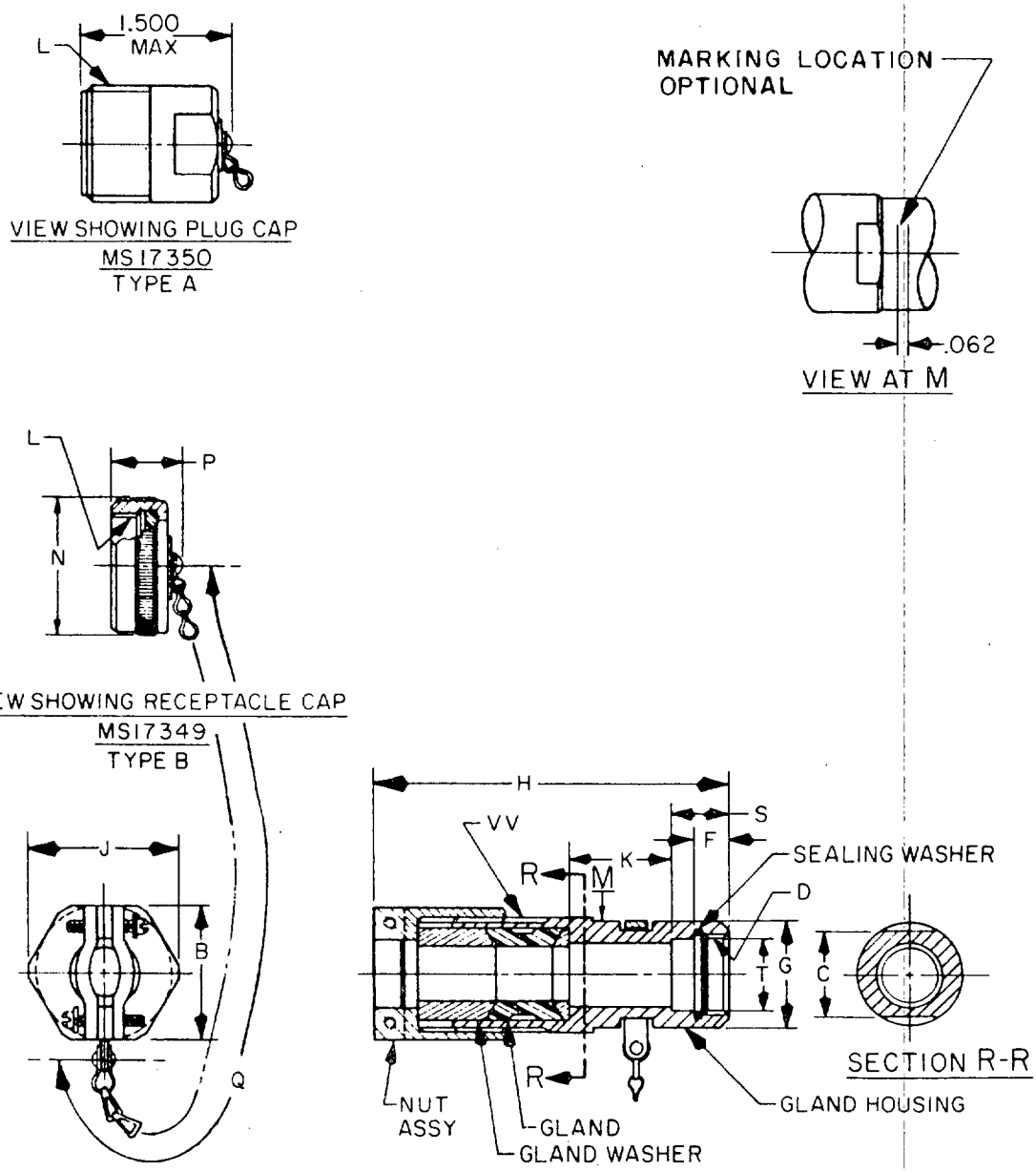
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

SIZE NO.	SHELL SIZE	CABLE RANGE MAX DIA	CABLE RANGE MIN DIA	B +.000 -.010	C +.000 -.010	D THREAD CLASS 2B-LH	E FREE LG MAX	F ±.016	G DIA +.010 -.020	H MAX	J ±.031	K +.015 -.025	L THREAD (PLATED) CLASS 2(A OR B)	N DIA MAX	P MAX	Q WITHIN ONE LINK
28	36	1.825	1.700	2.438	2.500	2.3125-16UNS	1.391	.479	2.562	5.354	3.171	1.344	2.5000-0.1P-0.2L-DS	2.719	0.980	6.00
29	36	1.984	1.859	2.438	2.500	2.3125-16 UNS	↑	.479	2.562	5.354	3.171	1.344	2.5000-0.1P-0.2L-DS	2.719	↑	↑
30	36	2.062	1.917	2.750	2.500	2.3125-16UNS	↑	.479	2.562	5.385	3.375	1.375	2.5000-0.1P-0.2L-DS	2.719	↑	↑
31	40	2.375	2.230	3.000	2.812	2.6250-16UN	↑	.667	2.875	5.682	3.625	1.421	2.7500-0.1P-0.2L-DS	2.969	↑	↑
32	↑	2.250	2.105	2.875	2.625	↑	↑	↑	↑	5.682	3.500	1.421	↑	↑	↑	↑
33	↓	2.145	2.000	2.750	2.625	↓	↓	↓	↓	5.620	3.375	1.422	↓	↓	↓	↓
34	40	2.062	1.917	2.750	2.625	2.6250-16UN	1.391	.667	2.875	5.620	3.375	1.422	2.7500-0.1P-0.2L-DS	2.969	.980	6.00

Configuration A. Style 1 cable sealing - Continued.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.



Configuration B. Style 2 cable sealing.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

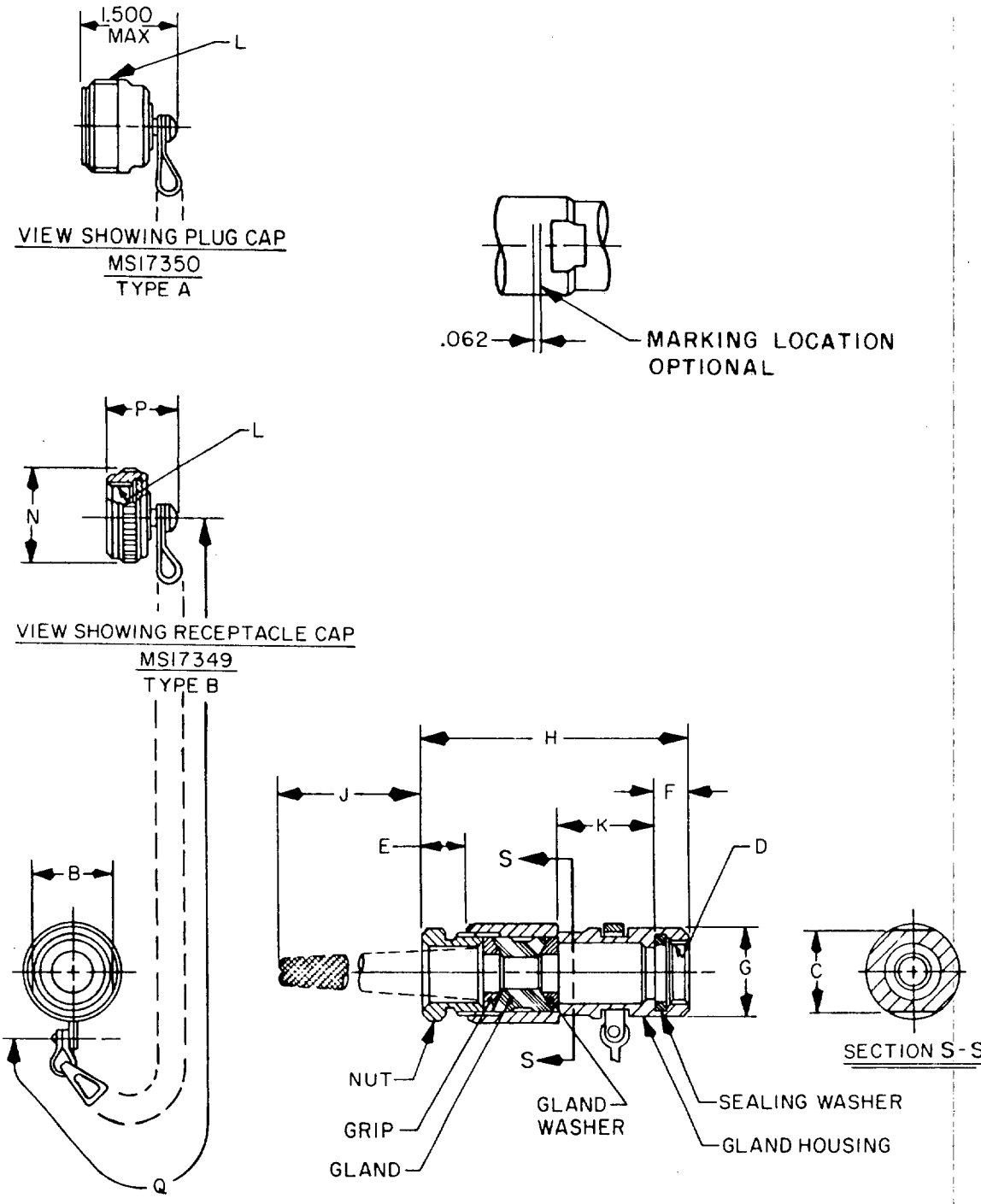
SIZE NO.	SHELL SIZE	CABLE RANGE MAX.	CABLE RANGE MIN.	B ±.005	C +.000 -.010	D THREAD CLASS 2B - LH	F ±.016	G DIA. +.010 -.020	H MAX.	J MAX.	K ±.010
1	12	..521	.321	1.219	.938	0.7500-20 UNEF	.479	.938	4.000	1.320	.950
2	14	.646	.446	1.344	1.062	0.8750-20 UNEF	↑	1.062	4.100	1.420	1.075
3	16	.765	.565	1.469	1.188	1.0000-20 UNEF	↓	1.188	4.800	1.600	1.200
4	18	.890	.690	1.625	1.312	1.1250-18 NEF	↓	1.312	4.900	1.750	1.325
5	20	1.015	.815	1.750	1.438	1.2500-18 NEF	↓	1.438	4.900	1.900	1.325
6	22	1.180	.980	2.000	1.562	1.3750-18 NEF	↓	1.562	4.900	2.190	1.325
7	24	1.430	1.230	2.250	1.812	1.6250-18 NEF	↓	1.812	5.000	2.475	1.450
8	24	1.230	1.030	2.250	1.812	1.6250-18 NEF	↓	1.812	5.000	2.475	1.450
9	28	1.670	1.430	2.500	2.062	1.8750-18 UN	↓	2.062	5.300	2.715	1.700
10	32	1.860	1.660	2.750	2.312	2.0625-16 N	↓	2.312	5.600	2.900	1.950
11	36	2.110	1.860	3.062	2.562	2.3125-16 N	.479	2.562	5.725	3.250	↑
12	40	2.400	2.200	3.437	2.875	2.6250-16 UN	.667	2.875	5.725	3.625	↓
13	40	2.200	2.000	3.437	2.875	2.6250-16 UN	.667	2.875	5.725	3.625	1.950

SIZE NO.	SHELL SIZE	L THREAD CLASS 2A/B	N DIA. MAX.	P MAX.	Q WITHIN ONE LINK	S ±.010	T DIA. ±.005	VV GLAND THREADS CLASS 2 - LH
1	12	0.8750-0.1P-0.2L-DS	1.094	.765	5.00	1.425	.600	1.0625-18 UNEF
2	14	1.0000-0.1P-0.2L-DS	1.219	.765	↑	1.425	.726	1.1875-18 NEF
3	16	1.1250-0.1P-0.2L-DS	1.344	.980	↓	1.605	.845	1.3125-18 NEF
4	18	1.2500-0.1P-0.2L-DS	1.469	↑	↓	↑	.970	1.4375-18 NEF
5	20	1.3750-0.1P-0.2L-DS	1.562	↓	5.00	↓	1.095	1.5625-18 NEF
6	22	1.5000-0.1P-0.2L-DS	1.688	↓	6.00	↓	1.220	1.8125-16 N
7	24	1.7500-0.1P-0.2L-DS	1.938	↓	↑	↓	1.470	2.0625-16 N
8	24	1.7500-0.1P-0.2L-DS	1.938	↓	↓	↓	1.470	2.0625-16 N
9	28	2.0000-0.1P-0.2L-DS	2.219	↓	↓	↓	1.713	2.3125-16 N
10	32	2.2500-0.1P-0.2L-DS	2.469	↓	↓	↓	1.900	2.5000-16 UN
11	36	2.5000-0.1P-0.2L-DS	2.719	↓	↓	↓	2.150	2.8750-16 UN
12	40	2.7500-0.1P-0.2L-DS	2.969	↓	↓	↓	2.460	3.1250-16 UN
13	40	2.7500-0.1P-0.2L-DS	2.969	.980	6.00	1.605	2.460	3.1250-16 UN

Configuration B. Style 2 cable sealing - Continued.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.



Configuration C. Style 3 cable sealing and gripping.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

SIZE NO	SHELL SIZE	CABLE RANGE MAX DIA	CABLE RANGE MIN DIA	B +.000 -.010	C +.000 -.010	D THREAD CLASS 2B-LH	E FREE LG MAX	F ±.016	G DIA +.010 -.020	H MAX	J APPROX FREE LENGTH	K +.015 -.025	L THREAD (PLATED) CLASS 2(A OR B)	N DIA MAX	P MAX	Q WITHIN ONE LINK
1	12	.530	.436	1.000	.812	.7500-20UNEF	.597	.479	.938	3.810	4.688	1.219	.8750-0.1P-0.2L-DS	1.094	.765	5.00
2	12	.500	.406	.875	.938	.7500-	.539	"	.938	3.248	4.344	.871	.8750-	1.094	"	"
3	12	.405	.316	1.000	.812	.7500-	.597		.938	3.810	3.688	1.219	.8750-	1.094		
4	14	.625	.560	1.062	1.125	.8750-	"		1.062	4.024	6.188	1.371	1.0000-	1.219		
5	14	.605	.511	1.000	1.062	.8750-			"	3.712	5.188	1.121	1.0000-	"		
6	14	.530	.430	1.000	1.062	.8750-				3.712	4.688	1.121	1.0000-			
7	14	.405	.316	1.000	1.062	.8750-20 UNEF			1.062	3.712	3.688	1.121	1.0000-	1.219	.765	
8	18	.828	.715	1.125	1.250	1.1250-18UNEF			1.312	4.087	6.688	1.343	1.2500-	1.469	.980	
9	20	1.055	.930	1.500	1.375	1.2500-			1.438	4.212	7.688	1.371	1.3750-	1.562	"	
10	20	1.000	.875	1.500	"	1.2500-			"	4.212	7.188	"	1.3750-	"		
11	20	.900	.787	1.250		1.2500-				4.150	7.188		1.3750-			
12	20	.750	.637	1.250	1.375	1.2500-			1.438	4.150	6.688		1.3750-	1.562		5.00
13	22	1.109	.984	1.500	1.625	1.3750-			1.562	4.212	7.688		1.5000-	1.688		6.00
14	22	1.000	.875	1.500	1.625	1.3750-			1.562	4.212	7.188	1.371	1.5000-	1.688		"
15	24	1.375	1.250	2.000	1.875	1.6250-			1.812	4.275	9.688	1.309	1.7500-	1.938		
16	24	1.310	1.185	1.750	"	1.6250-			"	"	8.688	1.372	1.7500-	"		
17	24	1.230	1.105	1.750		1.6250-					8.188	1.372	1.7500-			
18	24	1.180	1.055	1.750	1.875	1.6250-18UNEF			1.812	4.275	8.188	1.372	1.7500-	1.938		
19	28	1.531	1.406	2.000	2.062	1.8750-16UN			2.062	4.337	10.688	1.371	2.0000-	2.219		
20	28	1.445	1.320	2.000	2.062	1.8750-16UN	"		2.062	4.337	9.688	1.371	2.0000-	2.219		
21	28	1.375	1.250	2.000	2.062	1.8750-16UN	.597		2.062	4.337	9.688	1.371	2.0000-	2.219		
22	32	1.825	1.700	2.438	2.312	2.0625-16UNS	.710		2.312	4.704	13.688	1.375	2.2500-	2.469		
23	32	1.730	1.605	2.438	2.312	2.0625-	.710		2.312	4.704	13.688	1.375	2.2500-	2.469		
24	32	1.656	1.531	2.250	2.312	2.0625-	.597		2.312	4.399	12.688	1.370	2.2500-	2.469		
25	36	2.062	1.917	2.750	2.500	2.3125-	.710		2.562	4.704	14.188	1.375	2.5000-	2.719		
26	36	1.984	1.859	2.438	"	2.3125-	"		"	4.673	13.688	1.344	2.5000-	"		
27	36	1.900	1.775	"		2.3125-							2.5000-			
28	36	1.825	1.700			2.3125-							2.5000-			
29	36	1.730	1.605	2.438	2.500	2.3125-16UNS	.710	.479	2.562	4.673	13.688	1.344	2.5000-	2.719		
30	40	2.375	2.230	2.875	2.812	2.6250-16UN	.709	.667	2.875	5.000	18.188	1.421	2.7500-	2.969		
31	40	2.250	2.105	2.750	2.625	2.6250-	.709	"	"	5.000	17.188	1.421	2.7500-	"		
32	40	2.145	2.000	2.750	2.625	2.6250-	.710			4.939	14.188	1.422	2.7500-			
33	40	2.062	1.917	2.750	2.625	2.6250-16UN	.710	.667	2.875	4.939	14.188	1.422	2.7500-0.1P-0.2L-DS	2.969	.980	6.00

Configuration C. Style 3 cable sealing and gripping. - Continued.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.010	.25	1.600	40.64	2.625	66.68	3.678	93.42	4.337	110.16	4.959	125.96
.020	.51	1.688	42.88	2.715	68.96	3.712	94.28	4.399	111.73	5.000	127.00
.031	.79	1.750	44.45	2.875	73.02	3.810	96.77	4.489	114.02	5.021	127.53
.938	23.83	1.812	46.02	2.900	73.66	4.000	101.60	4.584	116.43	5.083	129.11
1.062	26.97	1.900	48.26	2.953	75.01	4.024	102.21	4.647	118.03	5.300	134.62
1.125	28.58	2.062	52.37	3.171	80.54	4.087	103.81	4.673	118.69	5.354	135.99
1.312	33.32	2.125	53.98	3.248	82.50	4.100	104.14	4.704	119.48	5.385	136.78
1.320	33.53	2.190	55.63	3.250	82.55	4.150	105.41	4.800	121.92	5.600	142.24
1.375	34.92	2.312	58.72	3.375	85.72	4.177	106.10	4.896	124.36	5.620	142.75
1.420	36.07	2.469	62.71	3.500	88.90	4.212	106.98	4.900	124.46	5.682	144.32
1.438	36.53	2.475	62.86	3.625	92.08	4.275	108.58	4.939	125.45	5.725	145.42
1.562	39.67	2.562	65.07								

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.016 (.41 mm).
4. Dimensions apply after plating.

FIGURE 1. Configurations and dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

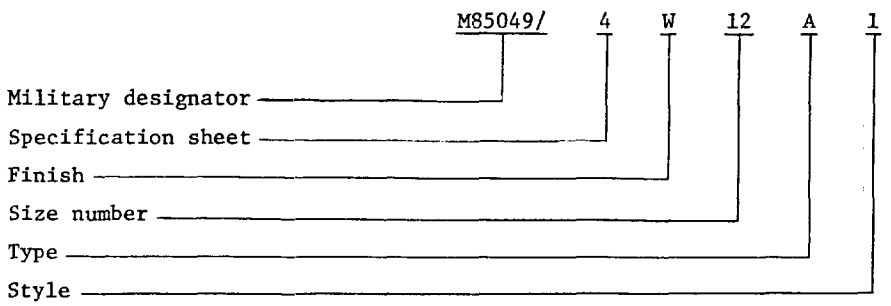
REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Accessories shall be designated type A and B.
- Type A is for use with MS 17344.
- Type B is for use with MS17343, MS17345, and MS17347.
The backshell, cap and chain is considered the complete accessory to be supplied.
Material and finish: See table I.
Connectors and their associated accessories shall be of the same finish.

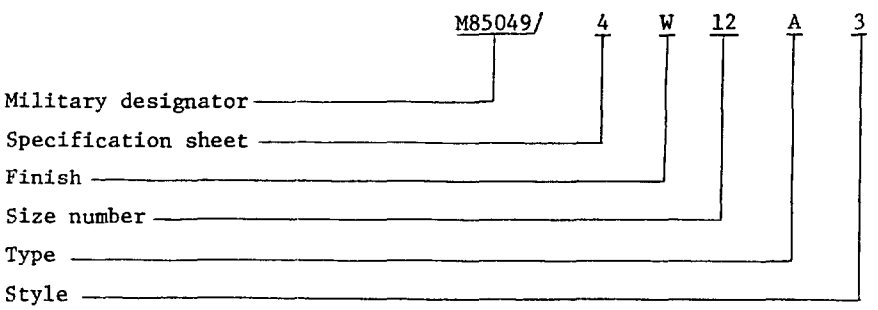
TABLE I. Material and finish.

Component	Material	Finish
Accessory body and cap	High grade aluminum alloy in accordance with MIL-C-85049	A, W
Chain	Corrosion resistant steel, Type II, Class 3, Trade number 8 of RR-C-271	Passivated
Gasket and gland	Rubber, in accordance with MIL-R-3065	

Part number: See examples below:
For styles 1 and 2:



For style 3:



Simplified assembly instructions shall be packaged with each accessory.

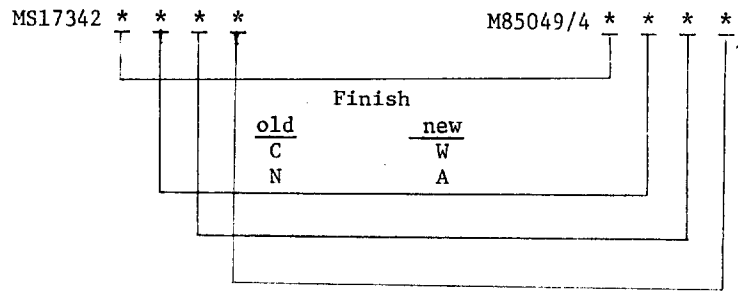
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

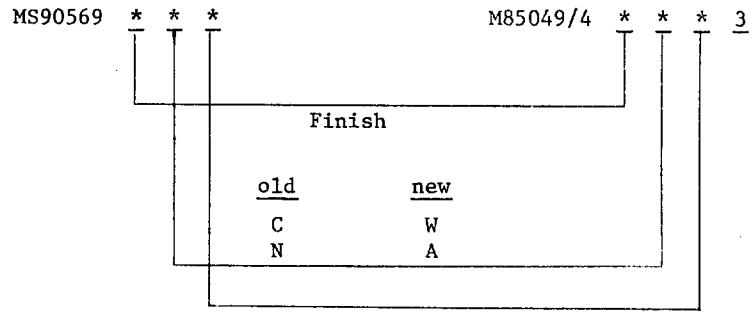
Application:
These accessories are for use with connectors specified on MS17343, MS17344, MS17345 and MS17347.
- Styles 1 and 3 are for use with classes R and C.
- Style 2 is for use with classes R, C and J.

Qualification:
See MIL-C-85049, category 1A.

QPL evaluating activity: Defense Electronics Supply Center, (DESC-E), Dayton, OH 45444.

Supersession data:





Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3067-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
SPACE AND NAVAL WARFARE SYSTEMS COMMAND
(SPAWAR 8111)
WASHINGTON, D.C. 203363

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:10Z
Check the source to verify that this is the current version before use.